field E is related to the current I flowing through the loop. For contour C, $V_{\text{emf}}^{\text{tr}}$ is related to $\mathbf{E}$ by

$$V_{\text{emf}}^{\text{tr}} = \oint_C \mathbf{E} \cdot d\mathbf{l}. \tag{6.10}$$

For $N = 1$ (a loop with one turn), equating Eqs. (6.8) and (6.10) gives

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}, \tag{6.11}$$

which is the integral form of Faraday's law given in **Table 6-1**. We should keep in mind that the direction of the contour C and the direction of $d\mathbf{s}$ are related by the right-hand rule.

By applying Stokes's theorem to the left-hand side of Eq. (6.11), we have

$$\int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{s} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}, \tag{6.12}$$

and in order for the two integrals to be equal for all possible choices of S, their integrands must be equal, which gives

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \qquad \textbf{(Faraday's law)} \tag{6.13}$$

This differential form of Faraday's law states that a time-varying magnetic field induces an electric field $\mathbf{E}$ whose curl is equal to the negative of the time derivative of $\mathbf{B}$. Even though the derivation leading to Faraday's law started out by considering the field associated with a physical circuit, Eq. (6.13) applies at any point in space, whether or not a physical circuit exists at that point.

Example 6-1: Inductor in a Changing Magnetic Field

An inductor is formed by winding N turns of a thin conducting wire into a circular loop of radius a. The inductor loop is in the x–y plane with its center at the origin and connected to a resistor R, as shown in **Fig. 6-3**. In the presence of a magnetic field $\mathbf{B} = B_0(\hat{\mathbf{y}}2 + \hat{\mathbf{z}}3)\sin\omega t$, where ω is the angular frequency, find

(a) the magnetic flux linking a single turn of the inductor,

(b) the transformer emf given that $N = 10$, $B_0 = 0.2$ T, $a = 10$ cm, and $\omega = 10^3$ rad/s,

(c) the polarity of $V_{\text{emf}}^{\text{tr}}$ at $t = 0$, and

(d) the induced current in the circuit for $R = 1$ kΩ (assume the wire resistance to be much smaller than R).

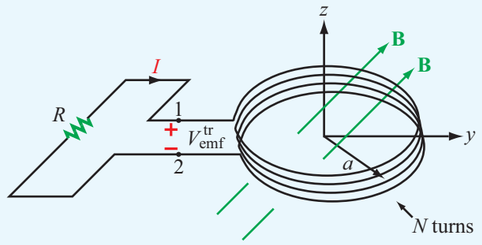


Figure 6-3 Circular loop with N turns in the x–y plane. The magnetic field is $\mathbf{B} = B_0(\hat{\mathbf{y}}2 + \hat{\mathbf{z}}3)\sin\omega t$ (Example 6-1).

Solution: (a) With $d\mathbf{s}$ defined upwards (along $+\hat{\mathbf{z}}$ direction), the magnetic flux linking each turn of the inductor is

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s} = \int_S [B_0(\hat{\mathbf{y}}2 + \hat{\mathbf{z}}3)\sin\omega t] \cdot \hat{\mathbf{z}}\, ds = 3\pi a^2 B_0 \sin\omega t.$$

(b) To find $V_{\text{emf}}^{\text{tr}}$, we can apply Eq. (6.8) or apply the general expression given by Eq. (6.6) directly. The latter approach gives

$$V_{\text{emf}}^{\text{tr}} = -N\frac{d\Phi}{dt} = -\frac{d}{dt}(3\pi N a^2 B_0 \sin\omega t) = -3\pi N\omega a^2 B_0 \cos\omega t.$$

Having chosen the direction of $d\mathbf{s}$ to be upwards, the polarity of $V_{\text{emf}}^{\text{tr}}$ is governed by the right-hand rule requiring the direction of the contour C in **Fig. 6-3** to pass across the gap from the positive terminal of $V_{\text{emf}}^{\text{tr}}$ to the negative terminal. That is,

$$V_{\text{emf}}^{\text{tr}} = V_1 - V_2 = -3\pi N\omega a^2 B_0 \cos\omega t.$$

For $N = 10$, $a = 0.1$ m, $\omega = 10^3$ rad/s, and $B_0 = 0.2$ T,

$$V_{\text{emf}}^{\text{tr}} = -188.5\cos 10^3 t \qquad \text{(V)}.$$

(c) The flux passing through each loop in the upward direction is $\Phi(t) = 3\pi a^2 B_0 \sin\omega t$. At $t = 0$, $\Phi(0) = 0$, but

$$\left.\frac{d\Phi}{dt}\right|_{t=0} = \left.3\pi a^2 \omega B_0 \cos\omega t_0\right|_{t=0} = 3\pi a^2 \omega B_0 > 0.$$

Hence, even though the flux itself is zero at $t = 0$, its time derivative is positive. Therefore, the flux is increasing at $t = 0$. Lenz's law requires the direction of the current I to be such that the flux it induces, Φ_{ind}, opposes the change in Φ. Since Φ is defined to be upwards, and it is increasing at $t = 0$,

磁场 E 与流过回路的电流 I 相关。对于轮廓 C，$V_{\text{emf}}^{\text{tr}}$ 与 E 的关系为

$$V_{\text{emf}}^{\text{tr}} = \oint_C \mathbf{E} \cdot d\mathbf{l}. \tag{6.10}$$

对于 N = 1 （一圈循环），等式：(6.8) 和 (6.10) 给出

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}, \tag{6.11}$$

这是表6-1中给出的法拉第定律的积分形式。我们应该记住，轮廓 C 的方向和 ds 的方向通过右手定则相关。

将斯托克斯定理应用于方程的左侧。 (6.11)，我们有

$$\int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{s} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}, \tag{6.12}$$

为了使两个积分对于 S 的所有可能选择都相等，它们的被积函数必须相等，这给出

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \qquad \textbf{(Faraday's law)} \tag{6.13}$$

法拉第定律的这种微分形式指出，随时间变化的磁场会感应出一个电场 E，其旋度等于 B 的时间导数的负值。尽管法拉第定律的推导是从考虑与物理电路相关的场开始的，方程 (6.13) 适用于空间中的任何一点，无论该点是否存在物理电路。

例 6-1： 变化磁场中的电感器

电感器是将一根细导线缠绕 N 匝，形成半径为 a 的圆环。电感环路位于 x-y 平面上，中心位于原点，并连接电阻器 R，如图 6-3 所示。在存在磁场 $\mathbf{B} = B_0(\hat{\mathbf{y}}2 + \hat{\mathbf{z}}3)\sin\omega t$ 的情况下，其中 ω 是角频率，找到

(a) 连接单匝电感器的磁通量，(b) 变压器电动势，假设 N = 10、$B_0 = 0.2$ T、a = 10 cm 和 $\omega = 10^3$ rad/s，(c) t = 0 时 $V_{\text{emf}}^{\text{tr}}$ 的极性，以及 (d) R 电路中的感应电流= 1 kΩ（假设导线电阻远小于 R)。

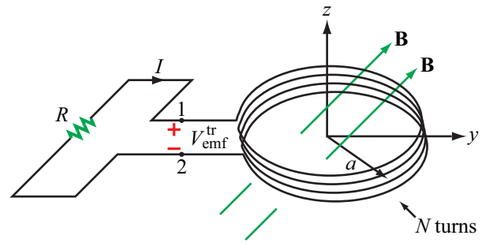


图 6-3 在 x-y 平面上有 N 匝的圆环。磁场为 $\mathbf{B} = B_0(\hat{\mathbf{y}}2 + \hat{\mathbf{z}}3)\sin\omega t$（例 6-1）。

解：(a) ds 向上定义（沿 $+\hat{\mathbf{z}}$ 方向），连接电感器每匝的磁通量为

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s} = \int_S [B_0(\hat{\mathbf{y}}2 + \hat{\mathbf{z}}3)\sin\omega t] \cdot \hat{\mathbf{z}}\, ds = 3\pi a^2 B_0 \sin\omega t.$$

(b) 要找到 $V_{\text{emf}}^{\text{tr}}$，我们可以应用方程： (6.8) 或应用式 (6.8) 给出的一般表达式。(6.6) 直接。后一种方法给出

$$V_{\text{emf}}^{\text{tr}} = -N\frac{d\Phi}{dt} = -\frac{d}{dt}(3\pi N a^2 B_0 \sin\omega t) = -3\pi N\omega a^2 B_0 \cos\omega t.$$

选择 ds 的方向为向上后，$V_{\text{emf}}^{\text{tr}}$ 的极性受右手法则控制，要求图 6-3 中轮廓 C 的方向穿过从 $V_{\text{emf}}^{\text{tr}}$ 正极端子到负极端子的间隙。那是，

$$V_{\text{emf}}^{\text{tr}} = V_1 - V_2 = -3\pi N\omega a^2 B_0 \cos\omega t.$$

对于 N = 10、a = 0.1 m、$\omega = 10^3$ rad/s 和 $B_0 = 0.2$ T，

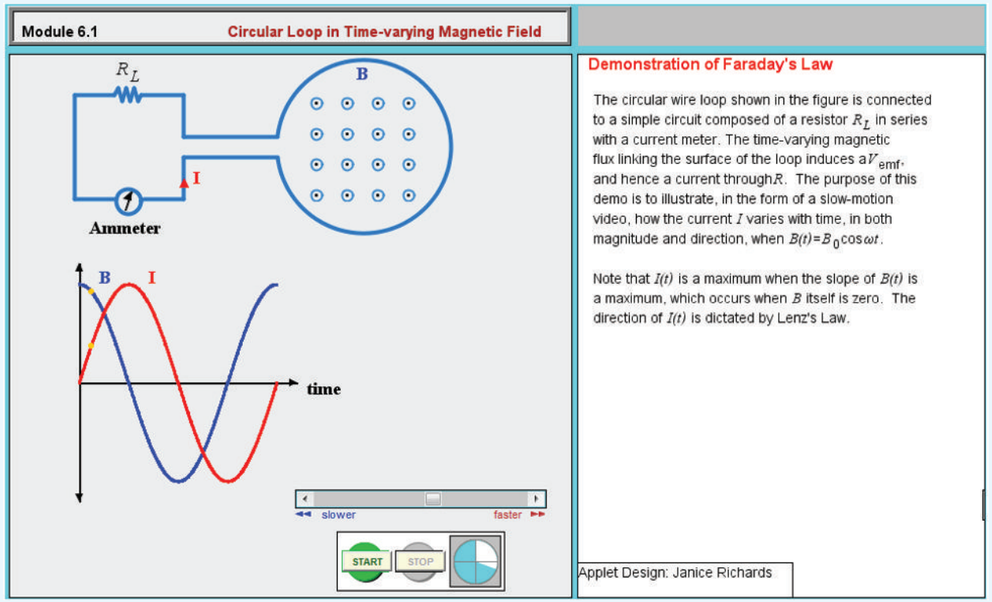
$$V_{\text{emf}}^{\text{tr}} = -188.5\cos 10^3 t \qquad \text{(V)}.$$

(c) 向上通过每个环路的通量为 $\Phi(t) = 3\pi a^2 B_0 \sin\omega t$。在 t = 0、$\Phi(0) = 0$ 时，但是

$$\left.\frac{d\Phi}{dt}\right|_{t=0} = \left.3\pi a^2 \omega B_0 \cos\omega t_0\right|_{t=0} = 3\pi a^2 \omega B_0 > 0.$$

因此，即使通量本身在 t = 0 处为零，其时间导数也是正的。 Therefore, the ux is increasing at t = 0. Lenz′ s law requires the direction of the current I to be such that the ux it induces, Φ_{ind}, opposes the change in Φ.由于 Φ 被定义为向上，并且它在 t = 0 处增加，

Module 6.1 Circular Loop in Time-Varying Magnetic Field Faraday's law of induction is demonstrated by simulating the current induced in a loop in response to the change in magnetic flux flowing through it.



Φ_{ind} should be downward through the loop in order to slow down the increase in Φ. Hence, I should be in the direction shown in **Fig. 6-3**.

Since I flows through a resistor from its positive voltage terminal to its negative voltage terminal, it follows that V_{12} should be negative at $t = 0$, which it is:

$$V_{\text{emf}}^{\text{tr}} = V_1 - V_2 = -188.5 \qquad \text{(V)}.$$

(d) The current I is given by

$$I = \frac{V_2 - V_1}{R} = \frac{188.5}{10^3}\cos 10^3 t = 0.19\cos 10^3 t \qquad \text{(A)}.$$

Exercise 6-1: For the loop shown in **Fig. 6-3**, what is $V_{\text{emf}}^{\text{tr}}$ if $\mathbf{B} = \hat{\mathbf{y}}B_0\cos\omega t$?

Answer: $V_{\text{emf}}^{\text{tr}} = 0$ because $\mathbf{B}$ is orthogonal to the loop's surface normal $d\mathbf{s}$. (See EM.)

Exercise 6-2: Suppose that the loop of Example 6-1 is replaced with a 10-turn square loop centered at the origin and having 20 cm sides oriented parallel to the x and y axes. If $\mathbf{B} = \hat{\mathbf{z}}B_0x^2\cos 10^3 t$ and $B_0 = 100$ T, find the current in the circuit.

Answer: $I = -133\sin 10^3 t$ (mA). (See EM.)

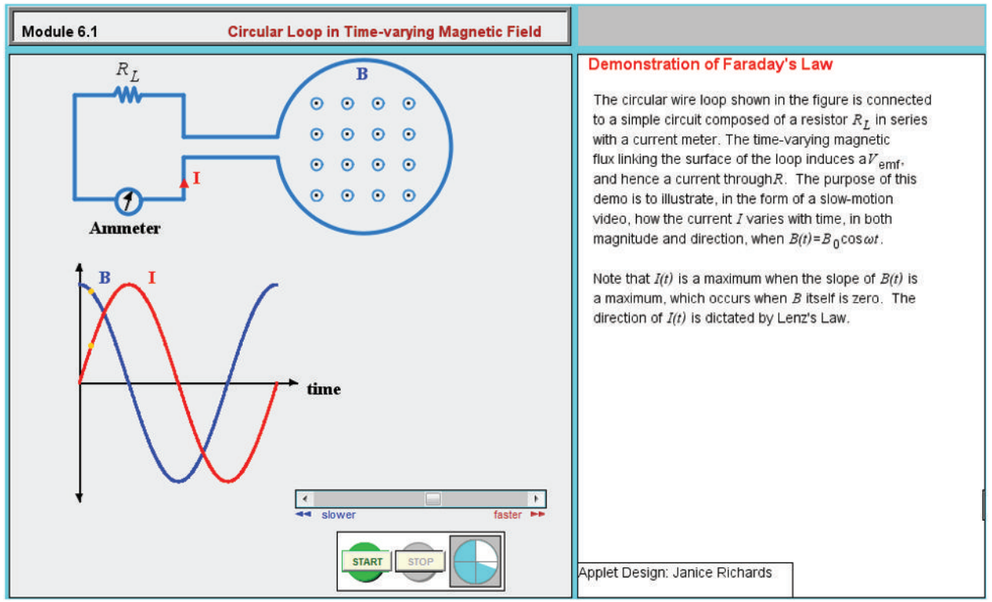
Example 6-2: Lenz's Law

Determine voltages V_1 and V_2 across the 2 Ω and 4 Ω resistors shown in **Fig. 6-4**. The loop is located in the x–y plane, its area is 4 m^2, the magnetic flux density is $\mathbf{B} = -\hat{\mathbf{z}}0.3t$ (T), and the internal resistance of the wire may be ignored.

Solution: The flux flowing through the loop is

$$\Phi = \int_S \mathbf{B}\cdot d\mathbf{s} = \int_S (-\hat{\mathbf{z}}0.3t)\cdot\hat{\mathbf{z}}\,ds = -0.3t\times 4 = -1.2t \qquad \text{(Wb)},$$

模块 6.1 时变磁场中的圆环 法拉第感应定律通过模拟环路中响应流过磁通量变化而感应的电流来演示。



Φ_{ind}应该通过循环向下，以减缓Φ的增加。因此，I 应该处于图 6-3 所示的方向。

由于 I 从电阻器的正电压端子流向负电压端子，因此 V_{12} 在 t = 0 处应为负值，即：

$$V_{\text{emf}}^{\text{tr}} = V_1 - V_2 = -188.5 \qquad \text{(V)}.$$

(d) 电流 I 由下式给出

$$I = \frac{V_2 - V_1}{R} = \frac{188.5}{10^3}\cos 10^3 t = 0.19\cos 10^3 t \qquad \text{(A)}.$$

练习 6-1：对于图 6-3 所示的循环，如果 $\mathbf{B} = \hat{\mathbf{y}}B_0\cos\omega t$，则 $V_{\text{emf}}^{\text{tr}}$ 是多少？

答案：$V_{\text{emf}}^{\text{tr}} = 0$，因为 B 与环的表面法线 ds 正交。（参见EM。）

练习 6-2：假设示例 6-1 中的环路被替换为以原点为中心、边长为 20 厘米、平行于 x 轴和 y 轴的 10 匝方形环路。如果 $\mathbf{B} = \hat{\mathbf{z}}B_0x^2\cos 10^3 t$ 且 $B_0 = 100$ T，求电路中的电流。

答案：$I = -133\sin 10^3 t$ (mA)。（参见EM。）

例 6-2：楞次定律

确定图 6-4 所示的 2 Ω 和 4 Ω 电阻两端的电压 V_1 和 V_2。环路位于x-y平面上，其面积为4m^2，磁通密度为$\mathbf{B} = -\hat{\mathbf{z}}0.3t$(T)，导线的内阻可以忽略不计。

解：流经环路的磁通为

$$\Phi = \int_S \mathbf{B}\cdot d\mathbf{s} = \int_S (-\hat{\mathbf{z}}0.3t)\cdot\hat{\mathbf{z}}\,ds = -0.3t\times 4 = -1.2t \qquad \text{(Wb)},$$

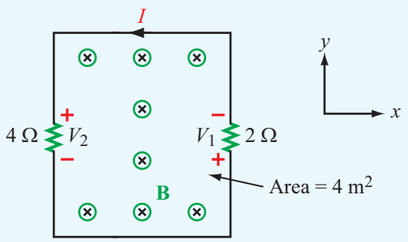


Figure 6-4 Circuit for Example 6-2.

and the corresponding transformer emf is

$$V_{\text{emf}}^{\text{tr}} = -\frac{d\Phi}{dt} = 1.2 \qquad \text{(V)}.$$

Since the magnetic flux through the loop is along the $-z$ direction (into the page) and increases in magnitude with time t, Lenz's law states that the induced current I should be in a direction such that the magnetic flux density $\mathbf{B}_{\text{ind}}$ it induces counteracts the direction of change of Φ. Hence, I has to be in the direction shown in the circuit because the corresponding $\mathbf{B}_{\text{ind}}$ is along the $+z$ direction in the region inside the loop area. This, in turn, means that V_1 and V_2 are positive voltages.

The total voltage of 1.2 V is distributed across two resistors in series. Consequently,

$$I = \frac{V_{\text{emf}}^{\text{tr}}}{R_1 + R_2} = \frac{1.2}{2+4} = 0.2 \text{ A},$$

and

$$V_1 = IR_1 = 0.2 \times 2 = 0.4 \text{ V},$$
$$V_2 = IR_2 = 0.2 \times 4 = 0.8 \text{ V}.$$

Concept Question 6-1: Explain Faraday's law and the function of Lenz's law.

Concept Question 6-2: Under what circumstances is the net voltage around a closed loop equal to zero?

Concept Question 6-3: Suppose the magnetic flux density linking the loop of **Fig. 6-4** (Example 6-2) is given by $\mathbf{B} = -\hat{\mathbf{z}}0.3e^{-t}$ (T). What would the direction of the current be, relative to that shown in **Fig. 6-4**, for $t \geq 0$?

6-3 The Ideal Transformer

The transformer shown in **Fig. 6-5(a)** consists of two coils wound around a common magnetic core. The primary coil has N_1 turns and is connected to an ac voltage source $V_1(t)$. The secondary coil has N_2 turns and is connected to a load resistor R_L. In an ideal transformer the core has infinite permeability ($\mu = \infty$), and the magnetic flux is confined within the core.

► The directions of the currents flowing in the two coils, I_1 and I_2, are defined such that, when I_1 and I_2 are both positive, the flux generated by I_2 is opposite to that generated by I_1. The transformer gets its name from the fact that it transforms currents, voltages, and impedances between its primary and secondary circuits, and vice versa. ◄

On the primary side of the transformer, the voltage source V_1 generates current I_1 in the primary coil, which establishes a flux Φ in the magnetic core. The flux Φ and voltage V_1 are

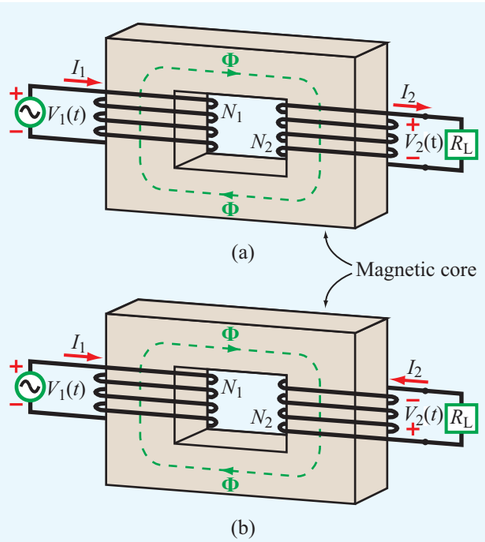


Figure 6-5 In a transformer, the directions of I_1 and I_2 are such that the flux Φ generated by one of them is opposite to that generated by the other. The direction of the secondary winding in (b) is opposite to that in (a), and so are the direction of I_2 and the polarity of V_2.

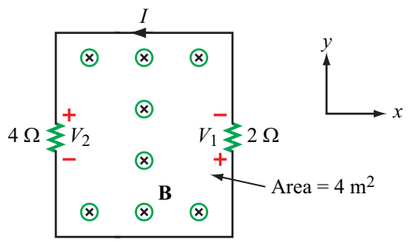


图 6-4 例 6-2 的电路。

6-3 理想的变压器

图 6-5(a) 所示的变压器由缠绕在公共磁芯上的两个线圈组成。初级线圈有N_1匝，并连接到交流电压源$V_1(t)$。次级线圈有 N_2 匝，并连接到负载电阻 R_L。在理想的变压器中，磁芯具有无限的磁导率 ($\mu = \infty$)，并且磁通量被限制在磁芯内。

► 两个线圈 I_1 和 I_2 中流动的电流方向被定义为，当 I_1 和 I_2 均为正时，I_2 产生的磁通与 I_1 产生的磁通相反。变压器因其在初级和次级电路之间变换电流、电压和阻抗而得名，反之亦然。◄

相应的变压器电动势为

$$V_{\text{emf}}^{\text{tr}} = -\frac{d\Phi}{dt} = 1.2 \qquad \text{(V)}.$$

由于穿过环路的磁通量沿着 $-z$ 方向（进入页面），并且其大小随时间 t 增加，因此楞次定律指出，感应电流 I 的方向应使其感应的磁通量密度 B_{ind} 抵消 Φ 的变化方向。因此，I 必须沿电路中所示的方向，因为相应的 B_{ind} 在环路区域内沿 $+z$ 方向。这又意味着 V_1 和 V_2 是正电压。

在变压器的初级侧，电压源 V_1 在初级线圈中产生电流 I_1，从而在磁芯中建立磁通 Φ。磁通 Φ 和电压 V_1 为 1.2 V 的总电压分布在两个串联电阻上。最后，

$$I = \frac{V_{\text{emf}}^{\text{tr}}}{R_1 + R_2} = \frac{1.2}{2+4} = 0.2 \text{ A},$$

和

$$V_1 = IR_1 = 0.2 \times 2 = 0.4 \text{ V},$$
$$V_2 = IR_2 = 0.2 \times 4 = 0.8 \text{ V}.$$

概念问题 6-1：解释法拉第定律和楞次定律的作用。

概念题6-2：什么情况下闭环周围的净电压为零？

概念问题 6-3：假设连接图 6-4（例 6-2）的回路的磁通密度由 $B = -\hat{z}0.3e^{-t}$ (T) 给出。相对于图 6-4 所示，对于 $t \geq 0$，电流的方向是什么？

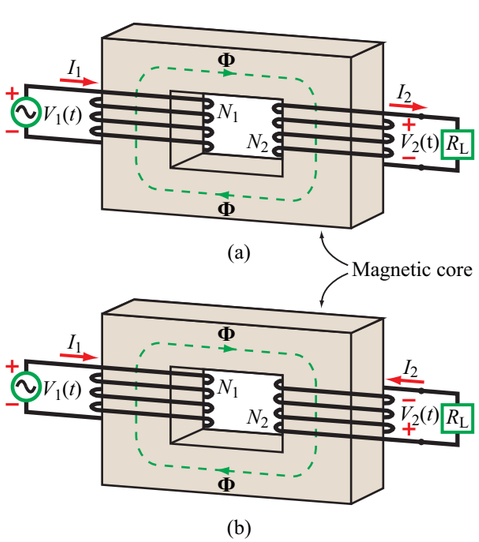


图 6-5 在变压器中，I_1 和 I_2 的方向使得其中一个产生的磁通 Φ 与另一个产生的磁通相反。 (b)中次级绕组的方向与(a)中的相反，I_2的方向和V_2的极性也相反。

related by Faraday's law:

$$V_1 = -N_1 \frac{d\Phi}{dt}. \tag{6.14}$$

A similar relation holds true on the secondary side:

$$V_2 = -N_2 \frac{d\Phi}{dt}. \tag{6.15}$$

The combination of Eqs. (6.14) and (6.15) gives

$$\frac{V_1}{V_2} = \frac{N_1}{N_2}. \tag{6.16}$$

In an ideal lossless transformer, all the instantaneous power supplied by the source connected to the primary coil is delivered to the load on the secondary side. Thus, no power is lost in the core, and

$$P_1 = P_2. \tag{6.17}$$

Since $P_1 = I_1 V_1$ and $P_2 = I_2 V_2$, and in view of Eq. (6.16), we have

$$\frac{I_1}{I_2} = \frac{N_2}{N_1}. \tag{6.18}$$

Thus, whereas the ratio of the voltages given by Eq. (6.16) is proportional to the corresponding turns ratio, the ratio of the currents is equal to the inverse of the turns ratio. If $N_1/N_2 = 0.1$, V_2 of the secondary circuit would be 10 times V_1 of the primary circuit, but I_2 would be only $I_1/10$.

The transformer shown in **Fig. 6-5(b)** is identical to that in **Fig. 6-5(a)** except for the direction of the windings of the secondary coil. Because of this change, the direction of I_2 and the polarity of V_2 in **Fig. 6-5(b)** are the reverse of those in **Fig. 6-5(a)**.

The voltage and current in the secondary circuit in **Fig. 6-5(a)** are related by $V_2 = I_2 R_L$. To the input circuit, the transformer may be represented by an equivalent input resistance R_{in}, as shown in **Fig. 6-6**, defined as

$$R_{in} = \frac{V_1}{I_1}. \tag{6.19}$$

Use of Eqs. (6.16) and (6.18) gives

$$R_{in} = \frac{V_2}{I_2}\left(\frac{N_1}{N_2}\right)^2 = \left(\frac{N_1}{N_2}\right)^2 R_L. \tag{6.20}$$

When the load is an impedance Z_L and V_1 is a sinusoidal source, the phasor-domain equivalent of Eq. (6.20) is

$$Z_{in} = \left(\frac{N_1}{N_2}\right)^2 Z_L. \tag{6.21}$$

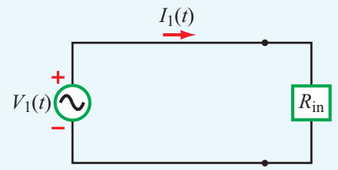


Figure 6-6 Equivalent circuit for the primary side of the transformer.

6-4 Moving Conductor in a Static Magnetic Field

Consider a wire of length l moving across a static magnetic field $\mathbf{B} = \hat{\mathbf{z}} B_0$ with constant velocity $\mathbf{u}$, as shown in **Fig. 6-7**. The conducting wire contains free electrons. From Eq. (5.3), the magnetic force $\mathbf{F}_m$ acting on a particle with charge q moving with velocity $\mathbf{u}$ in a magnetic field $\mathbf{B}$ is

$$\mathbf{F}_m = q(\mathbf{u} \times \mathbf{B}). \tag{6.22}$$

This magnetic force is equivalent to the electrical force that would be exerted on the particle by the electric field $\mathbf{E}_m$ given by

$$\mathbf{E}_m = \frac{\mathbf{F}_m}{q} = \mathbf{u} \times \mathbf{B}. \tag{6.23}$$

The field $\mathbf{E}_m$ generated by the motion of the charged particle is called a ***motional electric field*** and is in the direction perpendicular to the plane containing $\mathbf{u}$ and $\mathbf{B}$. For the wire shown in **Fig. 6-7**, $\mathbf{E}_m$ is along $-\hat{\mathbf{y}}$. The magnetic force acting

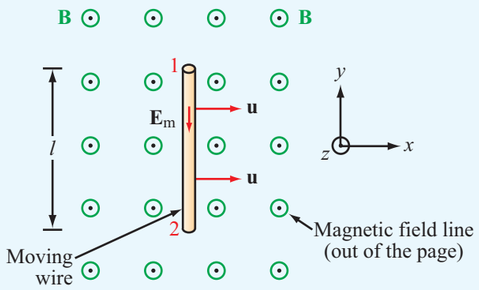


Figure 6-7 Conducting wire moving with velocity $\mathbf{u}$ in a static magnetic field.

由法拉第定律关联：

$$V_1 = -N_1 \frac{d\Phi}{dt}. \tag{6.14}$$

类似的关系也适用于次级侧：

$$V_2 = -N_2 \frac{d\Phi}{dt}. \tag{6.15}$$

等式的组合。 (6.14) 和 (6.15) 给出

$$\frac{V_1}{V_2} = \frac{N_1}{N_2}. \tag{6.16}$$

在理想的无损变压器中，连接到初级线圈的电源提供的所有瞬时功率都被传送到次级侧的负载。因此，核心不会损失任何能量，并且

$$P_1 = P_2. \tag{6.17}$$

由于 $P_1 = I_1V_1$ 和 $P_2 = I_2V_2$，并鉴于等式： (6.16) ，我们有

$$\frac{I_1}{I_2} = \frac{N_2}{N_1}. \tag{6.18}$$

因此，而方程给出的电压比。 (6.16) 与相应的匝数比成正比，电流之比等于匝数比的倒数。如果 $N_1/N_2 = 0.1$，则次级电路的 V_2 将是初级电路 V_1 的 10 倍，但 I_2 将仅为 $I_1/10$。

图6-5(b)所示的变压器与图6-5(a)所示的变压器相同，只是次级线圈的绕组方向不同。由于这种变化，图6-5(b)中I_2的方向和V_2的极性与图6-5(a)中的方向相反。

图6-5(a)中次级电路中的电压和电流之间的关系为$V_2 = I_2R_L$。对于输入电路来说，变压器可以用等效输入电阻R_{in}来表示，如图6-6所示，定义为

$$R_{in} = \frac{V_1}{I_1}. \tag{6.19}$$

使用等式。 (6.16) 和 (6.18) 给出

$$R_{in} = \frac{V_2}{I_2}\left(\frac{N_1}{N_2}\right)^2 = \left(\frac{N_1}{N_2}\right)^2 R_L. \tag{6.20}$$

当负载为阻抗 Z_L 且 V_1 为正弦源时，等式的相量域等效为： (6.20) 是

$$Z_{in} = \left(\frac{N_1}{N_2}\right)^2 Z_L. \tag{6.21}$$

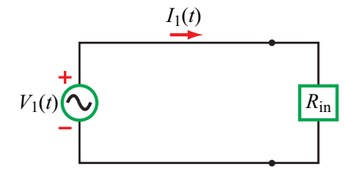


图 6-6 变压器初级侧的等效电路。

6-4 静磁场中移动的导体

考虑一根长度为 l 的导线以恒定速度 u 穿过静磁场 $B = \hat{z} B_0$，如图 6-7 所示。导线含有自由电子。从方程。 (5.3) 作用在磁场 B 中以速度 u 运动的带电荷 q 的粒子上的磁力 F_m 为

$$\mathbf{F}_m = q(\mathbf{u} \times \mathbf{B}). \tag{6.22}$$

该磁力相当于电场 E_m 施加在粒子上的电力，如下所示：

$$\mathbf{E}_m = \frac{\mathbf{F}_m}{q} = \mathbf{u} \times \mathbf{B}. \tag{6.23}$$

带电粒子运动产生的场 E_m 称为运动电场，其方向垂直于包含 u 和 B 的平面。对于图 6-7 所示的导线，E_m 沿 $-\hat{y}$ 方向。磁力作用

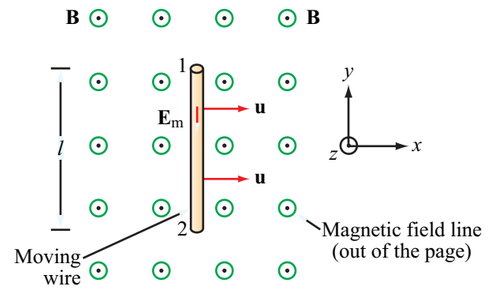


图 6-7 导线在静磁场中以速度 u 运动。

on the (negatively charged) electrons in the wire causes them to drift in the direction of $-\mathbf{E}_m$; that is, toward the wire end labeled 1 in **Fig. 6-7**. This, in turn, induces a voltage difference between ends 1 and 2, with end 2 being at the higher potential. The induced voltage is called a ***motional emf***, V_{emf}^m, and is defined as the line integral of $\mathbf{E}_m$ between ends 2 and 1 of the wire,

$$V_{emf}^m = V_{12} = \int_2^1 \mathbf{E}_m \cdot d\mathbf{l} = \int_2^1 (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}. \qquad (6.24)$$

For the conducting wire, $\mathbf{u} \times \mathbf{B} = \hat{\mathbf{x}}u \times \hat{\mathbf{z}}B_0 = -\hat{\mathbf{y}}uB_0$ and $d\mathbf{l} = \hat{\mathbf{y}}\, dl$. Hence,

$$V_{emf}^m = V_{12} = -uB_0 l. \qquad (6.25)$$

In general, if any segment of a closed circuit with contour C moves with a velocity $\mathbf{u}$ across a static magnetic field $\mathbf{B}$, then the induced motional emf is given by

$$V_{emf}^m = \oint_C (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}. \qquad \textbf{(motional emf)} \qquad (6.26)$$

▶ Only those segments of the circuit that cross magnetic field lines contribute to V_{emf}^m. ◀

Example 6-3: Sliding Bar

The rectangular loop shown in **Fig. 6-8** has a constant width l, but its length x_0 increases with time as a conducting bar slides with uniform velocity $\mathbf{u}$ in a static magnetic field $\mathbf{B} = \hat{\mathbf{z}}B_0 x$. Note that $\mathbf{B}$ increases linearly with x. The bar starts from $x = 0$ at $t = 0$. Find the motional emf between terminals 1 and 2 and the current I flowing through the resistor R. Assume that the loop resistance $R_i \ll R$.

Solution: This problem can be solved by using the motional emf expression given by Eq. (6.26) or by applying the general formula of Faraday's law. We now show that the two approaches yield the same result.

(a) Motional emf

The sliding bar, being the only part of the circuit that crosses the lines of the field $\mathbf{B}$, is the only part of contour 2341 that contributes to V_{emf}^m. Hence, at $x = x_0$, for example,

$$V_{emf}^m = V_{12} = V_{43} = \int_3^4 (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l} = \int_3^4 (\hat{\mathbf{x}}u \times \hat{\mathbf{z}}B_0 x_0) \cdot \hat{\mathbf{y}}\, dl = -uB_0 x_0 l.$$

The length of the loop is related to u by $x_0 = ut$. Hence,

$$V_{emf}^m = -B_0 u^2 lt \quad \text{(V)}. \qquad (6.27)$$

(b) Total emf

Since $\mathbf{B}$ is static, $V_{emf}^{tr} = 0$ and $V_{emf} = V_{emf}^m$ only. To verify that the same result can be obtained by the general form of Faraday's law, we evaluate the flux Φ through the surface of the loop. Thus,

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s} = \int_S (\hat{\mathbf{z}}B_0 x) \cdot \hat{\mathbf{z}}\, dx\, dy = B_0 l \int_0^{x_0} x\, dx = \frac{B_0 l x_0^2}{2}. \qquad (6.28)$$

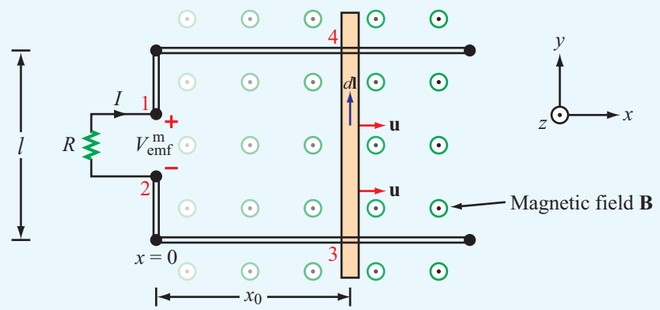


Figure 6-8 Sliding bar with velocity $\mathbf{u}$ in a magnetic field that increases linearly with x; that is, $\mathbf{B} = \hat{\mathbf{z}}B_0 x$ (Example 6-3).

导线中的（带负电的）电子会导致它们向 $-E_m$ 方向漂移；即朝向图 6-7 中标记为 1 的线端。这又会在 1 端和 2 端之间产生电压差，其中 2 端处于较高电势。感应电压称为动电动势 V_{emf}^m，定义为导线 2 端和 1 端之间 E_m 的线积分，

$$V_{emf}^m = V_{12} = \int_2^1 \mathbf{E}_m \cdot d\mathbf{l} = \int_2^1 (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}. \qquad (6.24)$$

对于导线，$u \times B = \hat{x}u \times \hat{z}B_0 = -\hat{y}uB_0$ 和 $dl = \hat{y}\, dl$。因此，

$$V_{emf}^m = V_{12} = -uB_0 l. \qquad (6.25)$$

一般来说，如果具有轮廓 C 的闭合电路的任何一段以速度 u 穿过静磁场 B，则感应运动电动势由下式给出

$$V_{emf}^m = \oint_C (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}. \qquad \textbf{(motional emf)} \qquad (6.26)$$

▶ 只有与磁场线交叉的电路部分才会产生 V_{emf}^m。 ◀

例 6-3：滑动条

图6-8所示的矩形环具有恒定的宽度l，但当导电棒在静磁场$B = \hat{z}B_0 x$中以匀速u滑动时，其长度x_0随着时间而增加。

请注意，B 随 x 线性增加。该条从 $t = 0$ 处的 $x = 0$ 开始。求端子 1 和 2 之间的运动电动势以及流过电阻器 R 的电流 I。假设环路电阻 $R_i \ll R$。

解决方案：这个问题可以通过使用方程给出的运动电动势表达式来解决。 (6.26) 或应用法拉第定律的一般公式。我们现在证明这两种方法产生相同的结果。

(a) 运动电动势

滑动条是电路中与场B的线交叉的唯一部分，也是轮廓2341中对V_{emf}^m有贡献的唯一部分。因此，例如，在 $x = x_0$ 处，

$$V_{emf}^m = V_{12} = V_{43} = \int_3^4 (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l} = \int_3^4 (\hat{\mathbf{x}}u \times \hat{\mathbf{z}}B_0 x_0) \cdot \hat{\mathbf{y}}\, dl = -uB_0 x_0 l.$$

循环的长度通过 $x_0 = ut$ 与 u 相关。因此，

$$V_{emf}^m = -B_0 u^2 lt \quad \text{(V)}. \qquad (6.27)$$

(b) 总电动势

由于 B 是静态的，因此仅 $V_{emf}^{tr} = 0$ 和 $V_{emf} = V_{emf}^m$。为了验证法拉第定律的一般形式是否可以得到相同的结果，我们评估了通过环路表面的通量 Φ。因此，

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s} = \int_S (\hat{\mathbf{z}}B_0 x) \cdot \hat{\mathbf{z}}\, dx\, dy = B_0 l \int_0^{x_0} x\, dx = \frac{B_0 l x_0^2}{2}. \qquad (6.28)$$

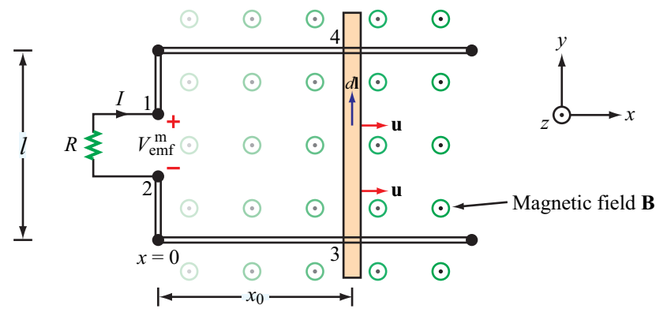


图 6-8 磁场中速度为 u 的滑动条，随 x 线性增加；即 $B = \hat{z}B_0 x$（例 6-3）。